

SOUMEE GHOSH

DATA SCIENTIST

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PROFESSIONAL SUMMARY:

Experienced **Data Scientist** specializing in leveraging advanced statistical methods, & cutting-edge AI/ML technologies to transform data into strategic advantages. Skilled in creating and optimizing Gen AI, LLMs, prompt engineering and ML models to drive actionable insights and deliver high-impact business solutions across diverse domains.

- Strong knowledge of **Machine Learning Algorithms**.
- Deep understanding of **Statistical Concepts**.
- Strong expertise in **Generative AI and LLMs**.
- Knowledge in **Deep Learning Algorithms**.
- Expertise in **Prompt Engineering Techniques**.
- Experience in **Natural Language Processing**
- Experience in **RAG implementation**.
- Experience in LLM -based Model Evaluation

Experience in working closely with clients in collaborative environments, swiftly adapting to new technologies and ensuring on time delivery of high-value solutions

SKILLS

- PROGRAMMING SKILLS: Python, SQL
- Machine Learning Algorithms: Linear & Logistic Regression, Decision Trees & Random Forests, Ensembling techniques like Boosting ,Bagging& Clustering algorithms, XG Boost, Support Vector Machines
- Natural Language Processing: Classical NLP techniques, Retrieval Augmented Generation (RAG), Generative AI, Large Language models (LLMs), Prompt Engineering , Chat GPT models, Elastic Search
- Deep Learning: ANN, CNN, RNN, LSTM
- Libraries used: NumPy, Pandas, Matplotlib, Plotly , Seaborn, TensorFlow, Keras, Statsmodels, NLTK, Spacy, Gensim,Scikit Learn, Spacy, Textacy, Open AI, Langchain, Sentence Transformers , Gradio.
- Platforms/IDEs: Jupyter notebook, VS code, Azure Databricks, Google Colab.
- Domain Knowledge: BFS, Retail.

EXPERIENCE

July 2022 - Present

Data Scientist – Cognizant Technology Solutions, Kolkata

Project: ASA Literature Review & Policy Note Generator

- Developed an AI-powered summary generator using RAG, LLMs, and Gen AI to process, analyse, and summarize multiple internal documents as well as user uploaded documents into cohesive research reports.
 - Optimized document retrieval by implementing Elasticsearch-based similarity search with document embeddings, significantly reducing processing time from 2 weeks to 2-3 seconds.
 - Automated contextual summarization by leveraging prompt engineering techniques to tailor document summaries based on user queries for specific topics.
 - Reduced manual effort by 99%, streamlining document preparation and improving user productivity by providing accurate, real-time project summaries with minimal human intervention.
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Project: AI-Driven Automated Report Generation

- Developed an innovative system to generate detailed annual reports by leveraging advanced prompt engineering techniques. The system effectively summarized and generated specific sections from multiple source documents, complemented by an auto-validation system to ensure content accuracy and quality.
- Crafted & refined section-specific prompts to effectively summarize and generate content from diverse source documents, facilitating accessibility & usability of lengthy documents leading to faster decision making.
- Implemented an automated validation framework to assess & enhance the accuracy & quality of generated reports.
- Leveraged validation feedback to iteratively improve prompt quality and overall report generation process, reducing the manual effort and time by almost 80 percent.

Project: AI-Auto Validator

- Developed and implemented an automated AI driven Validation system leveraging the G-eval framework and Gen AI to assess the accuracy and quality of AI-generated content.
- Streamlined the AI validation process, reducing manual review time by 70 % thereby increasing productivity and reliability of AI output evaluations.
- Incorporated a robust set of quality metrics and validation criteria to ensure comprehensive evaluation of AI content, including consistency, relevance and factual accuracy.

Project: Hotel Review Mapping.

- Developed a Hotel Review Mapping System and web application using Open AI's GPT-3.5 Turbo to process and accurately map over more than 1 lakh hotel review records (tabular data) to the correct names and addresses in the database significantly improving data integrity.
- Utilized Python's Fuzzy Matching library with GPT 3.5 Turbo to address challenges such as misspellings, abbreviations and incorrect address fields, ensuring accurate mapping of reviews despite data inconsistencies.
- Conducted thorough error analysis to resolve issues including zip code errors, duplicate hotel names and missing address fields, resulting in a more reliable system that reduces manual data correction efforts and enhances overall user experience.

Project: Advanced Time Series Analysis for Waste Management

- Developed a time series forecasting model for monthly waste recycled data at the reporting unit level.
- Conducted in-depth analysis to categorize reporting units based on operational status, addressing intermittent and lumpy time series data.
- Applied statistical models to active units with lumpy and intermittent data, outperforming conventional machine learning models.
- Utilized Light GBM and LSTM models for other data points, resulting in improved forecasting accuracy and operational efficiency.

Project: ISR Explorer

- Developed an AI system to extract and categorize key issues from Project Implementation Status reports, enhancing quick analysis.
- Generated issue summaries and assessed risk levels, aiding users in understanding and mitigating project concerns.
- Worked with both GPT-3.5 and GPT-4.0 models and optimized prompts by G-Eval framework validation to get optimized results, that helped in reducing errors.
- Achieved a 95% reduction in manual effort and facilitated quicker solution implementation.

Project: Advanced ML-Powered Anti Phishing Tool

- Engineered a cutting-edge anti-phishing system utilizing machine learning to classify URLs as malicious, suspicious or legitimate. Achieved a notable 89% accuracy with XGBoost classifier after hyperparameter tuning, compared to 80% accuracy with Random Forest, significantly enhancing web security.
- Performed extensive data handling and feature extraction developing robust lexical, keyword-based and host-based features from URL components.
- Applied grid search cross validation to identify and optimize hyperparameters for improved model performance.
- Delivered an effective solution for phishing detection, significantly improving the security and reliability of web interactions.

ACADEMICS:

- M.Sc. Economics, University of Calcutta, 2020-2022 (specialization: Econometrics, International Economics, Resource & Environmental Economics)
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AWARDS:

- Emerging Talent Award, Cognizant, 2024
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